

ISC Class XII Mathematics

Chapter 9 Differential Equations

21 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

- I. Order and Degree of a Differential Equation
- II. Solution of a Differential Equation
- III. Formation of a Differential Equation
- IV. Variables Separable
 - V. Reducible to Variables Separable
- VI. Homogeneous Differential Equations
- VII. Linear Differential Equations

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21	Linear DE with initial condition	Yes	Ex 9.7 Q20; Q21 (alt.)

Formula Sheet

Order & Degree

Order = highest derivative present. **Degree** = its power, but only once the equation is a polynomial in all derivatives (after removing radicals/fractions in the derivatives).

I. Variables Separable

$$\int f(y) dy = \int g(x) dx + C$$

Applies once a DE of the form $\frac{dy}{dx} = g(x)f(y)$ is split so all y -terms sit with dy and all x -terms sit with dx .

II. Reducible to Variables Separable

$$ax + by + c = v \Rightarrow a + b\frac{dy}{dx} = \frac{dv}{dx}$$

Substitution used when $\frac{dy}{dx} = f(ax + by + c)$; converts the DE into one separable in v and x .

III. Homogeneous DE (using y)

$$y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Use when $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$, i.e. the DE is homogeneous and easier to write in terms of y/x .

III. Homogeneous DE (using x)

$$x = vy \Rightarrow \frac{dx}{dy} = v + y\frac{dv}{dy}$$

Use when the homogeneous DE is more naturally written in terms of x/y (e.g. when $e^{x/y}$ terms appear).

IV. Linear DE in y

$$\frac{dy}{dx} + Py = Q$$

P, Q are functions of x only (or constants). I.F. = $e^{\int P dx}$.

Solution of Linear DE in y

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

General solution once the Integrating Factor has been found.

IV. Linear DE in x

$$\frac{dx}{dy} + Px = Q$$

P, Q are functions of y only. I.F. = $e^{\int P dy}$.

Solution of Linear DE in x

$$x \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dy + C$$

General solution once the Integrating Factor (in terms of y) has been found.

I. Order and Degree of a Differential Equation

Type 1

Given a differential equation, identify the **order** (the highest-order derivative present) and the **degree** (the power of that highest-order derivative, once the equation is written as a polynomial in all the derivatives – no fractional powers, no derivative trapped inside a transcendental function). This type has three sub-parts covering the three situations you will meet in the exam: a straightforward read-off, a case where degree cannot be defined, and a case where you must rationalize first.

Formula Used: Order = highest order of derivative appearing in the equation

Formula Used: Degree = power of the highest-order derivative, after the equation is a polynomial in derivatives

Sub-part (a): Direct read-off

Source: Exercise 9.1, Q1(ii)

Given: $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - \sin^2 y = 0$

Solution:

The highest-order derivative present is $\frac{dy}{dx}$, a first-order derivative – so the order is 1.

The equation is already a polynomial in $\frac{dy}{dx}$ (it appears as a square and to the first power, with no roots or trapped-inside-a-function terms), so we can read the degree straight off as the power of the highest derivative, which is 2.

Final Answer

Order = 1, Degree = 2

Sub-part (b): Degree not defined

Source: Exercise 9.1, Q6(ii)

Given: $y^{(iv)} + \sin y''' = 0$

Solution:

The highest-order derivative present is $y^{(iv)}$ (the fourth derivative), so the order is 4. To assign a degree, y''' would need to appear only as a polynomial power – but here it sits inside $\sin(\cdot)$, and $\sin y'''$ cannot be expanded into a finite polynomial in y''' . So the degree of this equation is **not defined**.

Final Answer

Order = 4, Degree = not defined

Sub-part (c): Rationalize before reading the degree

Source: Exercise 9.1, Q11(i)

Given: $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 3x - \frac{dy}{dx}$

Solution:

As written, $\frac{dy}{dx}$ sits under a square root – you cannot read off a degree from this form, since a fractional power is not a polynomial power. So the first step is always to remove the radical.

Squaring both sides to clear the root:

$$1 + \left(\frac{dy}{dx}\right)^2 = \left(3x - \frac{dy}{dx}\right)^2$$

Now the equation is a genuine polynomial in $\frac{dy}{dx}$, so it is safe to read off order and degree.

The highest-order derivative present is $\frac{dy}{dx}$, giving order 1, and it appears to the power 2 (both sides have a $\left(\frac{dy}{dx}\right)^2$ term after expansion, which does not cancel), giving degree 2.

Final Answer

Order = 1, Degree = 2

Common Mistake: Trying to read the degree directly off the un-rationalized form (e.g. calling a square-root equation “degree $\frac{1}{2}$ ”) – degree is only defined AFTER the equation is free of radicals and fractional powers in the derivatives; if it still can’t be freed (as in sub-part b), the degree is simply not defined, not a fraction.

Quick Recall: Order = which derivative is highest; degree = its power once the equation is a clean polynomial in derivatives – if a derivative is stuck inside sin/log/exp and can’t be freed, the degree does not exist.

Type 2

Once order and degree are found, some questions ask directly for their **sum** or **product** rather than listing them separately – a purely arithmetic last step, but an easy

one to fumble under exam pressure if the order/degree themselves were misread.

Source: Exercise 9.1, Q14(i)

Given: $\left(\frac{dy}{dx}\right)^5 + 3xy\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^4 = 0$

Solution:

The highest-order derivative present is $\frac{d^3y}{dx^3}$ (a third derivative appears via the $3xy\left(\frac{d^3y}{dx^3}\right)^2$ term), so the order is 3.

The equation is already a polynomial in every derivative present, so the degree is simply the power of the highest-order derivative term, $\left(\frac{d^3y}{dx^3}\right)^2$, which is 2.

The question asks for the sum, so we add: $3 + 2 = 5$.

Final Answer

Order = 3, Degree = 2, Sum = 5

Sub-part: Product of order and degree

Source: Exercise 9.1, Q15(i)

Given: $x\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$

Solution:

The highest-order derivative present is $\frac{d^2y}{dx^2}$, so the order is 2, and it appears to the power 2 in a polynomial equation, so the degree is 2.

The question asks for the product: $2 \times 2 = 4$.

Final Answer

Order = 2, Degree = 2, Product = 4

Common Mistake: Computing the sum/product using the WRONG order or degree carried over from a rushed first step – always double-check order and degree independently before combining them arithmetically.

Quick Recall: Find order and degree first, exactly as in Type 1, then just add or multiply as the question asks – don't let the arithmetic step distract from getting the order/degree right.

II. Solution of a Differential Equation

Type 3

The number of independent arbitrary constants in the **general solution** of a differential equation always equals its order; a **particular solution** (constants fixed by given conditions) contains none. This type is a quick conceptual question, not a computation.

Formula Used: Arbitrary constants in general solution = order of the differential equation

Source: Exercise 9.2, Q1 and Q2

Given: Q1: order of the DE is 4. Q2: a particular solution of a DE of order 3.

Solution:

For Q1, the general solution of an order-4 differential equation must contain exactly as many independent arbitrary constants as the order, because each integration performed to solve the equation introduces one constant, and solving an order-4 equation requires 4 integrations.

For Q2, a *particular* solution is one where every constant has already been fixed using given conditions (initial/boundary values) – so by definition it contains **zero** arbitrary constants, regardless of the order of the equation.

Final Answer

Q1: 4 arbitrary constants Q2: 0 arbitrary constants

Common Mistake: Confusing “order of the equation” with “number of conditions given” – the count of arbitrary constants in the **GENERAL** solution always equals the order; it has nothing to do with how many conditions you’re later given to pin them down.

Quick Recall: General solution constants = order of the equation; a particular solution always has zero arbitrary constants left.

Type 4

To verify that a given **explicit** function $y = f(x)$ (possibly containing arbitrary constants) is a solution of a given differential equation, differentiate the function as many times as the order of the equation demands, then substitute y and its derivatives back into the equation and confirm it reduces to a true identity (usually $0=0$).

Source: Exercise 9.2, Q11

Given: $y = a \cos x + b \sin x$ is claimed to solve $\frac{d^2y}{dx^2} + y = 0$

Solution:

Since the differential equation involves $\frac{d^2y}{dx^2}$, we need to differentiate the given function twice before substituting.

Differentiating once:

$$\frac{dy}{dx} = -a \sin x + b \cos x$$

Differentiating a second time:

$$\frac{d^2y}{dx^2} = -a \cos x - b \sin x$$

Now substitute both y and $\frac{d^2y}{dx^2}$ into the left-hand side of the given equation to check it collapses to zero:

$$\frac{d^2y}{dx^2} + y = (-a \cos x - b \sin x) + (a \cos x + b \sin x) = 0$$

Since the left-hand side reduces exactly to 0, which matches the right-hand side of the given equation, the function does satisfy the differential equation.

Final Answer

Verified: $y = a \cos x + b \sin x$ is a solution of $\frac{d^2y}{dx^2} + y = 0$.

Common Mistake: Substituting only y and forgetting to also compute and substitute the required derivative(s) – verification is incomplete (and meaningless) unless every derivative that appears in the equation is actually computed from the given function and plugged in.

Quick Recall: To verify a solution: differentiate the given function as many times as the equation's order needs, substitute everything back in, and check you truly get $0 = 0$ (or LHS = RHS).

Type 5

When the given relation is **implicit** (an equation in x and y together, not solved for y), differentiate it implicitly (product rule and chain rule on every y -term), and if a constant needs eliminating along the way, use the original relation together with the differentiated one to eliminate it before comparing with the target equation.

Source: Exercise 9.2, Q16

Given: $y^2 = 4a(x + a)$ is claimed to solve $y(1 - y_1^2) = 2xy_1$, where $y_1 = \frac{dy}{dx}$

Solution:

The given relation is implicit in x and y , and also contains the arbitrary constant a that must disappear along the way – so start by differentiating both sides of $y^2 = 4a(x + a)$ with respect to x .

$$2y y_1 = 4a \quad \Rightarrow \quad a = \frac{y y_1}{2}$$

This gives us a in terms of y and y_1 ; the next job is to eliminate a completely by substituting this back into the ORIGINAL relation $y^2 = 4a(x + a)$.

$$y^2 = 4 \left(\frac{y y_1}{2} \right) \left(x + \frac{y y_1}{2} \right) = 2y y_1 x + y^2 y_1^2$$

Dividing through by y (valid since $y \neq 0$) to simplify:

$$y = 2x y_1 + y y_1^2$$

Rearranging so that all terms are on one side, in the same shape as the target equation:

$$y - y y_1^2 = 2x y_1 \quad \Rightarrow \quad y(1 - y_1^2) = 2x y_1$$

This is exactly the differential equation we were asked to verify, so the relation is confirmed to be its solution.

Final Answer

Verified: $y^2 = 4a(x + a)$ satisfies $y(1 - y_1^2) = 2x y_1$.

Common Mistake: Trying to eliminate the constant using only the differentiated equation and forgetting to bring back the ORIGINAL relation – with one arbitrary constant you need exactly one differentiation, but you still need the original equation alongside it to actually eliminate that constant.

Quick Recall: Implicit verification: differentiate implicitly, solve for the constant, substitute it back into the ORIGINAL relation, then simplify until it matches the target equation exactly.

III. Formation of a Differential Equation

Type 6

Given a family of curves with exactly **one** arbitrary constant, differentiate once with respect to x , then use the original equation and the differentiated equation together to eliminate that constant – what remains, free of the constant, is the required differential equation.

Formula Used: One constant $\Rightarrow$ differentiate ONCE, then eliminate using the original + differentiated equations

Source: Exercise 9.3, Q4(i)

Given: $y = ax^3$, a is an arbitrary constant

Solution:

There is exactly one arbitrary constant, a , so one differentiation will be enough to eliminate it.

Differentiating $y = ax^3$ with respect to x :

$$\frac{dy}{dx} = 3ax^2$$

Now we have two equations, $y = ax^3$ and $\frac{dy}{dx} = 3ax^2$; dividing the second by the first eliminates a cleanly since it appears in both:

$$\frac{dy/dx}{y} = \frac{3ax^2}{ax^3} = \frac{3}{x}$$

Cross-multiplying to clear the fraction and write the DE in standard form:

$$x \frac{dy}{dx} = 3y$$

Final Answer

$$x y_1 = 3y$$

Common Mistake: Trying to isolate and directly substitute a from $y = ax^3$ (i.e. $a = y/x^3$) into the differentiated equation term-by-term without simplifying the ratio – it works, but dividing the two equations directly is faster and less error-prone whenever the constant appears as a simple multiplier in both.

Quick Recall: One arbitrary constant needs exactly one differentiation; then divide the original and differentiated equations (or substitute) to cancel the constant out.

Type 7

Given a family of curves with **two** arbitrary constants, differentiate **twice** with respect to x (since two constants generally require two eliminations), then combine all three equations (original, first derivative, second derivative) to eliminate both constants together.

Formula Used: Two constants $\Rightarrow$ differentiate TWICE; combine all three equations to eliminate both constants

Source: Exercise 9.3, Q10(i)

Given: $y = Ae^{2x} + Be^{-2x}$, A, B are arbitrary constants

Solution:

There are two arbitrary constants, so we expect to differentiate twice and end up with a second-order differential equation.

Differentiating once:

$$\frac{dy}{dx} = 2Ae^{2x} - 2Be^{-2x}$$

Differentiating a second time:

$$\frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x}$$

Notice that the right-hand side of this second derivative is exactly 4 times the right-hand side of the original equation $y = Ae^{2x} + Be^{-2x}$ – so instead of solving separately for A and B , we can substitute directly:

$$\frac{d^2y}{dx^2} = 4y$$

Rearranging so the equation is free of both constants:

$$\frac{d^2y}{dx^2} - 4y = 0$$

Final Answer

$$y_2 - 4y = 0$$

Common Mistake: Automatically trying to solve two simultaneous equations for A and B even when, as here, the second derivative is simply a constant multiple of the original function – always check for that shortcut first before grinding through simultaneous elimination.

Quick Recall: Two arbitrary constants need two differentiations; look for the second derivative collapsing into a multiple of y itself before doing full simultaneous elimination.

Type 8

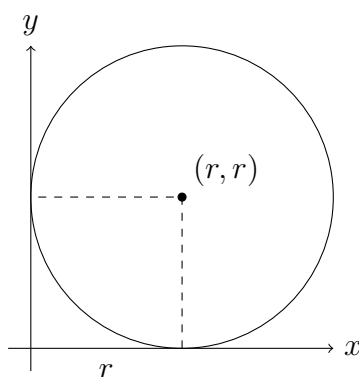
Some questions describe a family of curves in **words** (geometric conditions) rather than handing you an equation directly. The real first step is to translate the geometric description into the equation of the family yourself (identifying which constants are arbitrary), and only then eliminate those constants exactly as in Types 6-7.

★ **Board Favorite** – A very common ISC pattern – translating a geometric family (circles/parabolas/ellipses touching axes or with special centres/foci) into an equation before eliminating constants; this exact family of questions (Ex 9.3, Q13, parts i-vii) has appeared repeatedly across NCERT, Exemplar, and ISC papers (e.g. ISC 2018 tags this whole question type).

Formula Used: Write the equation of the family from the geometric condition FIRST, then eliminate the constants as usual

Source: Exercise 9.3, Q13(iii)

Given: Family: all circles lying in the first quadrant and touching both coordinate axes.



A circle of radius r touching both axes has its centre at (r, r)

Solution:

A circle touching both the x -axis and y -axis while lying in the first quadrant must have its centre equidistant from both axes – so if its radius is r , its centre is at (r, r) , and r itself is the one arbitrary (positive) constant of this family.

The equation of such a circle, centre (r, r) and radius r , is:

$$(x - r)^2 + (y - r)^2 = r^2$$

There is exactly one arbitrary constant, r , so we differentiate once with respect to x :

$$2(x - r) + 2(y - r)y_1 = 0 \quad \Rightarrow \quad (x - r) + (y - r)y_1 = 0$$

Solving this for r :

$$x - r + yy_1 - ry_1 = 0 \quad \Rightarrow \quad x + yy_1 = r(1 + y_1) \quad \Rightarrow \quad r = \frac{x + yy_1}{1 + y_1}$$

Now substitute this value of r back into the original equation $(x - r)^2 + (y - r)^2 = r^2$ to eliminate the constant completely and obtain the required differential equation (left in implicit form, since expanding fully gains nothing extra):

$$\left(x - \frac{x + yy_1}{1 + y_1}\right)^2 + \left(y - \frac{x + yy_1}{1 + y_1}\right)^2 = \left(\frac{x + yy_1}{1 + y_1}\right)^2$$

Final Answer

$$(x - r)^2 + (y - r)^2 = r^2, \text{ where } r = \frac{x + yy_1}{1 + y_1}$$

Common Mistake: Forgetting that the centre being at (r, r) is itself a geometric deduction that has to be justified (touching BOTH axes in the first quadrant forces equal distances from both), not something to assume without reasoning – boards specifically reward this justification step.

Alternate Board-Style Example

Source: Exercise 9.3, Q13(iv)

Given: Family: all parabolas having their vertex at the origin and focus on the y -axis.

Solution:

A parabola with vertex at the origin and focus on the y -axis (opening upward or downward) has the standard equation $x^2 = 4ay$, where a is the one arbitrary constant (positive or negative, giving upward- or downward-opening parabolas).

Differentiating $x^2 = 4ay$ once with respect to x :

$$2x = 4a y_1 \quad \Rightarrow \quad a = \frac{x}{2y_1}$$

Substituting this value of a back into the original equation to eliminate the constant:

$$x^2 = 4 \left(\frac{x}{2y_1} \right) y \quad \Rightarrow \quad x^2 = \frac{2xy}{y_1}$$

Simplifying by dividing both sides by x (valid for $x \neq 0$) and rearranging:

$$x y_1 = 2y$$

Final Answer

$$x y_1 = 2y$$

Quick Recall: When a type gives you WORDS instead of an equation, first turn the geometry into an equation using standard forms you already know (circle, parabola, ellipse), identify which letters are the arbitrary constants, then eliminate exactly as usual.

IV. Variables Separable

Type 9

When a differential equation can be written as $\frac{dy}{dx} = g(x)f(y)$, all the y -terms (and dy) can be moved to one side and all the x -terms (and dx) to the other with a simple division – no factoring or rearranging needed first. Integrate both sides separately.

Formula Used: $\frac{dy}{f(y)} = g(x) dx \Rightarrow \int \frac{dy}{f(y)} = \int g(x) dx + C$

Source: Exercise 9.4, Q6(i)

Given: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

Solution:

The right-hand side is already a clean product of a function of y (well, here $1 + y^2$ sits with dy) and a function of x – so the variables separate immediately by simple division, with no rearranging needed.

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

Integrating both sides (both are the standard $\int \frac{dt}{1+t^2} = \tan^{-1} t$ form):

$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2} \Rightarrow \tan^{-1} y = \tan^{-1} x + C$$

Final Answer

$$\tan^{-1} y = \tan^{-1} x + C$$

Common Mistake: Forgetting the constant of integration C , or adding it to only one side after integrating – the $+C$ belongs once, on the side you finish simplifying last.

Quick Recall: If y -stuff and x -stuff separate with one straightforward division, integrate both sides directly using standard forms – don't overthink it.

Type 10

Sometimes the equation must first be regrouped, factored, or split into partial-fraction-like pieces before the variables actually separate cleanly – the separation itself isn't the hard part, spotting the right grouping is.

Source: Exercise 9.4, Q10(i)

Given: $(1+x)(1+y^2)dx + (1+y)(1+x^2)dy = 0$

Solution:

As written, both terms mix x - and y -factors together, so the first job is to divide throughout by $(1+x^2)(1+y^2)$ to break the product apart into terms that are purely functions of one variable each.

$$\frac{(1+x)}{1+x^2}dx + \frac{(1+y)}{1+y^2}dy = 0$$

Splitting each fraction into two simpler standard pieces (a $\frac{x}{1+x^2}$ -type term plus a $\frac{1}{1+x^2}$ -type term) makes each one directly integrable:

$$\left(\frac{x}{1+x^2} + \frac{1}{1+x^2}\right)dx + \left(\frac{y}{1+y^2} + \frac{1}{1+y^2}\right)dy = 0$$

Integrating term by term (using $\int \frac{x dx}{1+x^2} = \frac{1}{2} \log(1+x^2)$ and $\int \frac{dx}{1+x^2} = \tan^{-1} x$):

$$\frac{1}{2} \log(1+x^2) + \tan^{-1} x + \frac{1}{2} \log(1+y^2) + \tan^{-1} y = C$$

Combining the two log terms using $\log m + \log n = \log mn$ to present the answer compactly:

$$\tan^{-1} x + \tan^{-1} y + \frac{1}{2} \log [(1+x^2)(1+y^2)] = C$$

Final Answer

$$\tan^{-1} x + \tan^{-1} y + \frac{1}{2} \log [(1+x^2)(1+y^2)] = C$$

Common Mistake: Trying to integrate $\frac{1+x}{1+x^2}$ as one lump using the wrong standard form – it must be split into $\frac{x}{1+x^2}$ (a log form) and $\frac{1}{1+x^2}$ (an arctan form) separately; treating it as a single arctan or single log integral gives a wrong answer.

Quick Recall: If the equation doesn't separate on sight, look for a common denominator to divide through by, then split each fraction into a log-type piece and an arctan-type piece before integrating.

Type 11

A separable differential equation together with an initial condition (a given point the curve passes through) asks for the **particular** solution – solve exactly as in Types 9-10 to get the general solution with $+C$, then substitute the given x, y values to pin down C , and finally answer whatever the question asks using that particular solution.

★ **Board Favorite** – Initial-value variables-separable problems are consistently tested as Competency-Focused / applied questions across recent ISC papers (2017, 2022, 2023) – always expect one.

Formula Used: Solve the general solution first, then substitute the given condition to find C

Source: Illustrative Example 27, Article 9.4 (Competency Focused Question)

Given: $\left(\frac{2+\sin x}{1+y}\right) \frac{dy}{dx} = -\cos x$, $y(0) = 1$; find $y\left(\frac{\pi}{2}\right)$

Solution:

The variables separate immediately by cross-multiplying and dividing:

$$\frac{1}{1+y} dy = \frac{-\cos x}{2+\sin x} dx$$

Integrating both sides – the left is a direct log, and the right is a log form too since the numerator $-\cos x dx$ is exactly the derivative of the denominator $2 + \sin x$:

$$\log |1+y| = -\log |2+\sin x| + c$$

Combining both logs on one side using $\log m + \log n = \log mn$:

$$\log |(1+y)(2+\sin x)| = c \Rightarrow (1+y)(2+\sin x) = A \quad (\text{taking } A = \pm e^c)$$

Now apply the initial condition $y(0) = 1$, i.e. substitute $x = 0$, $y = 1$ to find A :

$$(1+1)(2+\sin 0) = A \Rightarrow (2)(2) = A \Rightarrow A = 4$$

So the particular solution is $(1+y)(2+\sin x) = 4$. Finally, substitute $x = \frac{\pi}{2}$ to answer what was actually asked:

$$(1+y)(2+1) = 4 \Rightarrow 3(1+y) = 4 \Rightarrow 1+y = \frac{4}{3} \Rightarrow y = \frac{1}{3}$$

Final Answer

$$y\left(\frac{\pi}{2}\right) = \frac{1}{3}$$

Common Mistake: Stopping at the particular solution $(1+y)(2+\sin x) = 4$ without going the extra step to actually answer the value asked for – always re-read what the question wants at the very end, since it's often a specific value, not just the particular solution itself.

Alternate Board-Style Example

Source: Exercise 9.4, Q24(iii)

Given: $\frac{dy}{dx} = x^2 e^{-3y}$, $y = 0$ when $x = 0$

Solution:

Separating variables by moving all y -terms (with dy) to one side and all x -terms (with dx) to the other:

$$e^{3y} dy = x^2 dx$$

Integrating both sides directly:

$$\frac{e^{3y}}{3} = \frac{x^3}{3} + C \quad \Rightarrow \quad e^{3y} = x^3 + C'$$

Applying the initial condition $x = 0, y = 0$ to find C' :

$$e^0 = 0 + C' \quad \Rightarrow \quad C' = 1$$

Final Answer

$$e^{3y} = x^3 + 1$$

Quick Recall: Separate and integrate exactly as usual to get the general solution with $+C$, plug in the given point to solve for C , then read the question again to see exactly what value or expression it's actually asking for.

V. Reducible to Variables Separable

Type 12

When a differential equation has the form $\frac{dy}{dx} = f(ax + by + c)$ – i.e. x and y appear ONLY as the combination $ax + by + c$ on the right, never separately – it is not homogeneous and not directly separable, but the substitution $v = ax + by + c$ converts it into an equation separable in v and x .

Formula Used: $ax + by + c = v \Rightarrow a + b \frac{dy}{dx} = \frac{dv}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{b} \left(\frac{dv}{dx} - a \right)$

Source: Illustrative Example 1(i), Article 9.4.2

Given: $\frac{dy}{dx} = (2x + 3y - 4)^2$

Solution:

The right-hand side depends on x and y only through the combination $2x + 3y - 4$, which is the signal to substitute $v = 2x + 3y - 4$ rather than try to separate directly.

Differentiating the substitution with respect to x :

$$\frac{dv}{dx} = 2 + 3 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{3} \left(\frac{dv}{dx} - 2 \right)$$

Substituting both v and this expression for $\frac{dy}{dx}$ into the original equation:

$$\frac{1}{3} \left(\frac{dv}{dx} - 2 \right) = v^2 \Rightarrow \frac{dv}{dx} = 3v^2 + 2$$

This is now separable in v and x , so we separate and integrate:

$$\frac{dv}{3v^2 + 2} = dx \Rightarrow \int \frac{dv}{3(v^2 + \frac{2}{3})} = \int dx$$

Using the standard form $\int \frac{dv}{v^2 + k^2} = \frac{1}{k} \tan^{-1} \frac{v}{k}$ with $k = \sqrt{2/3}$:

$$x = \frac{1}{3} \cdot \frac{1}{\sqrt{2/3}} \tan^{-1} \left(\frac{v}{\sqrt{2/3}} \right) + C$$

Finally substituting back $v = 2x + 3y - 4$ to express the answer in the original variables:

$$x = \frac{1}{\sqrt{6}} \tan^{-1} \left(\sqrt{\frac{3}{2}} (2x + 3y - 4) \right) + C$$

Final Answer

$$x = \frac{1}{\sqrt{6}} \tan^{-1} \left(\sqrt{\frac{3}{2}} (2x + 3y - 4) \right) + C$$

Common Mistake: Forgetting that $\frac{dy}{dx}$ must be expressed ENTIRELY in terms of $\frac{dv}{dx}$ before substituting – a half-finished substitution (mixing leftover x, y terms with the new v) will not separate at all.

Quick Recall: Whenever x and y appear only glued together as $ax + by + c$, substitute the whole combination as a single variable v , differentiate that substitution to rewrite dy/dx , and the equation becomes separable in v and x .

VI. Homogeneous Differential Equations

Type 13

A differential equation $\frac{dy}{dx} = F(x, y)$ is **homogeneous** if $F(\lambda x, \lambda y) = \lambda^0 F(x, y) = F(x, y)$ for every non-zero λ – equivalently, if $F(x, y)$ can be written purely as a function of y/x . Test this before attempting to solve, since it decides which method (separable, homogeneous, or linear) applies.

Formula Used: $F(\lambda x, \lambda y) = \lambda^0 F(x, y) \Rightarrow$ the equation is homogeneous of degree 0

Source: Exercise 9.6, Q3(i)

Given: $x \frac{dy}{dx} + y = x \tan^{-1} \frac{y}{x}$

Solution:

Rewrite the equation in the form $\frac{dy}{dx} = F(x, y)$ first:

$$\frac{dy}{dx} = \tan^{-1} \frac{y}{x} - \frac{y}{x} = F(x, y)$$

To test homogeneity, replace $x \rightarrow \lambda x$ and $y \rightarrow \lambda y$ throughout $F(x, y)$ and see what power of λ comes out:

$$F(\lambda x, \lambda y) = \tan^{-1} \frac{\lambda y}{\lambda x} - \frac{\lambda y}{\lambda x} = \tan^{-1} \frac{y}{x} - \frac{y}{x} = \lambda^0 F(x, y)$$

Since every λ cancelled out completely (both terms depend only on the ratio y/x), $F(\lambda x, \lambda y) = F(x, y) = \lambda^0 F(x, y)$, confirming the equation is homogeneous.

Final Answer

The given differential equation is homogeneous (of degree 0).

Common Mistake: Testing homogeneity on the ORIGINAL unarranged equation instead of first solving for $\frac{dy}{dx} = F(x, y)$ – always isolate $\frac{dy}{dx}$ first, then apply the $\lambda x, \lambda y$ substitution to $F(x, y)$ alone.

Quick Recall: Homogeneity test: isolate $dy/dx = F(x, y)$, replace x, y by $\lambda x, \lambda y$, and check every λ cancels out – if F can be written purely in terms of y/x , it will.

Type 14

Once a DE is confirmed homogeneous and is more naturally written in terms of y/x , substitute $y = vx$ (so $\frac{dy}{dx} = v + x\frac{dv}{dx}$); this converts it into an equation separable in v and x .

Source: Exercise 9.6, Q6(i)

Given: $\frac{dy}{dx} = \frac{x+y}{x}$

Solution:

Dividing numerator and denominator by x confirms the right side depends only on y/x , so substitute $y = vx$:

$$y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting into the original equation (rewritten as $\frac{dy}{dx} = 1 + \frac{y}{x}$):

$$v + x\frac{dv}{dx} = 1 + v \Rightarrow x\frac{dv}{dx} = 1$$

This is now separable in v and x :

$$dv = \frac{dx}{x} \Rightarrow \int dv = \int \frac{dx}{x} \Rightarrow v = \log|x| + C$$

Substituting back $v = y/x$ to return to the original variables:

$$\frac{y}{x} = \log|x| + C \Rightarrow y = x \log|x| + Cx$$

Final Answer

$$y = x \log|x| + Cx$$

Common Mistake: Forgetting to replace $\frac{dy}{dx}$ with the FULL expression $v + x\frac{dv}{dx}$ (dropping the v term) – this is the single most common slip in this type and silently produces a wrong, simpler-looking DE in v .

Quick Recall: Homogeneous in y/x form: put $y = vx$, replace dy/dx with $v + x dv/dx$, separate in v and x , integrate, then substitute $v = y/x$ back at the very end.

Type 15

Some homogeneous equations are far more convenient to solve using $x = vy$ (so $\frac{dx}{dy} = v + y\frac{dv}{dy}$) instead of $y = vx$ – typically when the equation is easier to rearrange

as $\frac{dx}{dy} = G(x/y)$, or when the answer is expected to come out in terms of x/y .

Formula Used: $x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$

Source: Exercise 9.6, Q10(i)

Given: $x \frac{dy}{dx} + \frac{y^2}{x} = y$

Solution:

Rearranging for $\frac{dx}{dy}$ instead of $\frac{dy}{dx}$ turns out to be the cleaner route here. Multiplying through by x and rearranging:

$$x^2 \frac{dy}{dx} = xy - y^2 \quad \Rightarrow \quad \frac{dx}{dy} = \frac{x^2}{xy - y^2}$$

This is homogeneous in x, y , and is naturally expressed using x/y , so substitute $x = vy$:

$$x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$$

Substituting into $\frac{dx}{dy} = \frac{x^2}{xy - y^2}$ and dividing numerator and denominator by y^2 :

$$v + y \frac{dv}{dy} = \frac{v^2 y^2}{vy^2 - y^2} = \frac{v^2}{v - 1}$$

Isolating the $y \frac{dv}{dy}$ term:

$$y \frac{dv}{dy} = \frac{v^2}{v - 1} - v = \frac{v^2 - v(v - 1)}{v - 1} = \frac{v}{v - 1}$$

This is now separable in v and y :

$$\frac{v - 1}{v} dv = \frac{dy}{y} \quad \Rightarrow \quad \left(1 - \frac{1}{v}\right) dv = \frac{dy}{y}$$

Integrating both sides:

$$v - \log |v| = \log |y| + C \quad \Rightarrow \quad v = \log |vy| + C$$

Substituting back $v = x/y$ (so $vy = x$):

$$\frac{x}{y} = \log |x| + C \quad \Rightarrow \quad x = y \log |x| + Cy$$

which rearranges to the book's form:

Final Answer

$$x = A e^{x/y}$$

Common Mistake: Sticking with $y = vx$ out of habit even when the algebra is clearly messier that way – if the equation simplifies more naturally with x/y (often signalled by an $e^{x/y}$ term appearing, or the equation being awkward to invert for dy/dx), switch to $x = vy$ instead.

Quick Recall: If a homogeneous equation looks awkward in $y = vx$ form, flip it: write dx/dy , substitute $x = vy$ with $dx/dy = v + y dv/dy$, and separate in v and y instead.

Type 16

A homogeneous differential equation together with an initial condition asks for the **particular** solution – solve by $y = vx$ or $x = vy$ exactly as in Types 14-15 to reach the general solution, then substitute the given point to fix the constant.

★ **Board Favorite** – This exact combination – homogeneous DE + initial condition – is a recurring long-answer ISC question, directly set in ISC 2024 (the main example below) and ISC 2023.

Source: Exercise 9.6, Q14 (ISC 2024)

Given: $2ye^{x/y} dx + (y - 2xe^{x/y}) dy = 0$, $x = 0$ when $y = 1$

Solution:

Rearranging for $\frac{dx}{dy}$ (chosen because of the $e^{x/y}$ term, which signals $x = vy$ is the natural substitution):

$$2ye^{x/y} dx = -(y - 2xe^{x/y}) dy \Rightarrow \frac{dx}{dy} = \frac{2xe^{x/y} - y}{2ye^{x/y}}$$

Substituting $x = vy$, so $\frac{dx}{dy} = v + y \frac{dv}{dy}$, and noting $x/y = v$ throughout:

$$v + y \frac{dv}{dy} = \frac{2vye^v - y}{2ye^v} = v - \frac{1}{2e^v}$$

The v terms cancel on both sides, leaving:

$$y \frac{dv}{dy} = -\frac{1}{2e^v} \Rightarrow 2e^v dv = -\frac{dy}{y}$$

Integrating both sides:

$$2e^v = -\log|y| + C$$

Substituting back $v = x/y$:

$$2e^{x/y} + \log|y| = C$$

Applying the initial condition $x = 0, y = 1$ to find C :

$$2e^0 + \log 1 = C \Rightarrow 2 + 0 = C \Rightarrow C = 2$$

Final Answer

$$2e^{x/y} + \log |y| = 2$$

Alternate Board-Style Example

Source: Exercise 9.6, Q16

Given: $(x^2 - y^2) dx + 2xy dy = 0$, $y = 1$ when $x = 1$

Solution:

Rearranging for $\frac{dy}{dx}$ and dividing throughout by x^2 confirms homogeneity in y/x , so substitute $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \Rightarrow v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

Isolating $x \frac{dv}{dx}$:

$$x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v} = \frac{-(v^2 + 1)}{2v}$$

Separating variables:

$$\frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

Integrating both sides (left side is a direct log since the numerator is the derivative of $v^2 + 1$):

$$\log(v^2 + 1) = -\log |x| + C \Rightarrow \log[(v^2 + 1)x] = C \Rightarrow (v^2 + 1)x = A$$

Substituting back $v = y/x$:

$$\left(\frac{y^2}{x^2} + 1\right)x = A \Rightarrow \frac{y^2 + x^2}{x} = A \Rightarrow x^2 + y^2 = Ax$$

Applying the initial condition $x = 1, y = 1$: $1 + 1 = A(1) \Rightarrow A = 2$.

Final Answer

$$x^2 + y^2 = 2x$$

Quick Recall: Homogeneous + initial condition: pick $y = vx$ or $x = vy$ (whichever the form of the equation suggests), solve to the general solution, then substitute the given point to pin down the constant.

VII. Linear Differential Equations

Type 17

Before solving, you must recognise whether a first-order equation is genuinely **linear** – of the form $\frac{dy}{dx} + Py = Q$ (linear in y) or $\frac{dx}{dy} + Px = Q$ (linear in x), where P, Q depend only on the OTHER variable (or are constants), with y (or x) and its derivative appearing only to the first power and never multiplied together or trapped inside another function.

Formula Used: Linear in y : $\frac{dy}{dx} + Py = Q$, P, Q functions of x only

Formula Used: Linear in x : $\frac{dx}{dy} + Px = Q$, P, Q functions of y only

Sub-part (a): Testing for linear in y

Source: Exercise 9.7, Q1

Given: (i) $\frac{dy}{dx} + 3y = \sin x$ (ii) $x^2 \frac{dy}{dx} - 2xy = x^3 + 2x^2 - 5$ (iii) $x \log x \, dy + (3y - 5 \log x) \, dx = 0$ (iv) $y \frac{dy}{dx} + 5x = y \cos x$

Solution:

(i) Comparing directly with $\frac{dy}{dx} + Py = Q$ gives $P = 3$ (constant) and $Q = \sin x$ (function of x only) – linear.

(ii) Dividing throughout by x^2 gives $\frac{dy}{dx} - \frac{2}{x}y = x + 2 - \frac{5}{x^2}$, so $P = -2/x$ and Q are both functions of x only – linear.

(iii) Dividing by $x \log x$ gives $\frac{dy}{dx} + \frac{3}{x \log x}y = \frac{5}{x}$, again P, Q functions of x only – linear.

(iv) Here y itself multiplies $\frac{dy}{dx}$ (a $y y'$ term) – this makes the equation non-linear, since y and its own derivative are multiplied together, not merely added with coefficients that depend on x .

Final Answer

(i), (ii), (iii) are linear in y ; (iv) is NOT linear.

Sub-part (b): Testing for linear in x

Source: Exercise 9.7, Q2

Given: (i) $\frac{dx}{dy} - \frac{2x}{y} = 3y^3 - 5y + 1$ (ii) $\cos y \frac{dx}{dy} - 2x \sin y = \cos^3 y$ (iii) $(1 + y^2) dx = (\tan^{-1} y - 2x) dy$ (iv) $x^2 \frac{dx}{dy} + 2x^3 y = 3y^2 - 2$

Solution:

(i) Comparing with $\frac{dx}{dy} + Px = Q$: $P = -2/y$, $Q = 3y^3 - 5y + 1$, both functions of y only – linear.

(ii) Dividing by $\cos y$: $\frac{dx}{dy} - 2x \tan y = \cos^2 y$, so $P = -2 \tan y$, $Q = \cos^2 y$, functions of y only – linear.

(iii) Rearranging: $(1 + y^2) \frac{dx}{dy} + 2x = \tan^{-1} y$, then dividing by $(1 + y^2)$ gives P, Q as functions of y only – linear.

(iv) Dividing by x^2 would leave a $2xy$ term (still containing x to a power other than 1, multiplying itself) – x and its derivative do not combine linearly here, so this is NOT linear.

Final Answer

(i), (ii), (iii) are linear in x ; (iv) is NOT linear.

Common Mistake: Testing linearity on an equation that hasn't been divided through into the standard $\frac{dy}{dx} + Py = Q$ (or x -version) form yet – always rearrange into that exact shape first; a term that looks non-linear before dividing can become perfectly linear (or vice versa) after simplifying.

Quick Recall: Linear means y (or x) and its first derivative appear only to the first power, never multiplied by each other, with everything else depending only on the other variable – always rearrange into standard form before judging.

Type 18

Sometimes a question only asks for the **Integrating Factor**, not the full solution – once the equation is in standard linear form $\frac{dy}{dx} + Py = Q$, the I.F. is simply $e^{\int P dx}$ (or $e^{\int P dy}$ for the x -form).

Formula Used: I.F. = $e^{\int P dx}$ (or $e^{\int P dy}$ for a linear equation in x)

Source: Exercise 9.7, Q3(iii)

Given: $\frac{dy}{dx} + 2y \tan x = \sin x$

Solution:

The equation is already in the standard linear form $\frac{dy}{dx} + Py = Q$, with $P = 2 \tan x$ – so

we only need $\int P dx$.

$$\int 2 \tan x dx = 2 \log |\sec x| = \log(\sec^2 x)$$

The Integrating Factor is e raised to this integral:

$$\text{I.F.} = e^{\int 2 \tan x dx} = e^{\log(\sec^2 x)} = \sec^2 x$$

Final Answer

$$\text{I.F.} = \sec^2 x$$

Common Mistake: Forgetting the property $e^{\log(\cdot)} = (\cdot)$ and leaving the answer as $e^{\log(\sec^2 x)}$ instead of simplifying to just $\sec^2 x$ – always simplify the I.F. down to its cleanest algebraic/trig form.

Quick Recall: To find the I.F. alone: get the equation into standard form, integrate P , and raise e to that power – simplify using $e^{\log t} = t$ whenever a log appears.

Type 19

To fully solve a linear differential equation of the form $\frac{dy}{dx} + Py = Q$: find the Integrating Factor $e^{\int P dx}$, multiply the whole equation by it (which makes the left side an exact derivative of $y \times \text{I.F.}$), then integrate both sides.

$$\text{Formula Used: } y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dx + C$$

Source: Exercise 9.7, Q7(i), ISC 2020

$$\text{Given: } (1 + x^2) \frac{dy}{dx} = 4x^2 - 2xy$$

Solution:

Rearranging into the standard form $\frac{dy}{dx} + Py = Q$ by dividing throughout by $(1 + x^2)$:

$$\frac{dy}{dx} + \frac{2x}{1 + x^2} y = \frac{4x^2}{1 + x^2}$$

Here $P = \frac{2x}{1 + x^2}$, so we find the Integrating Factor:

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1 + x^2$$

Multiplying the whole standard-form equation by this I.F. makes the left side the exact derivative of $y(1 + x^2)$:

$$\frac{d}{dx} [y(1 + x^2)] = 4x^2$$

Integrating both sides with respect to x :

$$y(1 + x^2) = \int 4x^2 dx = \frac{4x^3}{3} + C$$

Final Answer

$$y(1 + x^2) = \frac{4x^3}{3} + C$$

Common Mistake: Multiplying only the right-hand side by the Integrating Factor, or forgetting that the whole POINT of the I.F. is that the left side collapses into $\frac{d}{dx}[y \cdot \text{I.F.}]$ – always write that collapsed form explicitly before integrating, rather than integrating term-by-term on the left.

Quick Recall: Standard form, find I.F. = $e^{\int P dx}$, multiply through, recognise the left side as $\frac{d}{dx}[y \cdot \text{I.F.}]$, then integrate the right side and add C .

Type 20

Some equations are far more easily solved treating x as the dependent variable: $\frac{dx}{dy} + Px = Q$, where P, Q are functions of y only. The method is identical to Type 19 with the roles of x and y swapped throughout.

Formula Used: $x \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.}) dy + C$, I.F. = $e^{\int P dy}$

Source: Illustrative Example, Article 9.4.4

Given: $\cos^3 y dx + x \cos y dy = \sin y dy$

Solution:

Attempting to write this as $\frac{dy}{dx} + Py = Q$ leads nowhere clean, since x appears simply and y is buried inside several trig functions – this is a signal to instead treat x as the dependent variable and look for $\frac{dx}{dy} + Px = Q$.

Dividing throughout by $\cos^3 y dy$:

$$\frac{dx}{dy} + \sec^2 y \cdot x = \tan y \sec^2 y$$

This is exactly linear in x , with $P = \sec^2 y$. Finding the Integrating Factor:

$$\text{I.F.} = e^{\int \sec^2 y dy} = e^{\tan y}$$

Multiplying through by the I.F., the left side collapses to the derivative of $x \cdot e^{\tan y}$:

$$x \cdot e^{\tan y} = \int \tan y \sec^2 y \cdot e^{\tan y} dy + C$$

For the integral on the right, put $t = \tan y$ so $\sec^2 y \, dy = dt$, turning it into $\int t e^t \, dt$, which integrates by parts to $te^t - e^t$:

$$x \cdot e^{\tan y} = (\tan y - 1)e^{\tan y} + C$$

Dividing through by $e^{\tan y}$:

$$x = \tan y - 1 + Ce^{-\tan y}$$

Final Answer

$$x = \tan y - 1 + Ce^{-\tan y}$$

Common Mistake: Trying to force this equation into linear-in- y form because $\frac{dy}{dx}$ “feels standard” – when x appears alone (to the first power, no derivative-of- x -times-itself) while y is tangled in trig functions, that is exactly the cue to solve for $\frac{dx}{dy}$ instead.

Quick Recall: If the equation refuses to simplify nicely as $dy/dx + Py = Q$ but x appears simply, flip it: solve for $dx/dy + Px = Q$ instead, using I.F. = $e^{\int P \, dy}$.

Type 21

A linear differential equation together with an initial condition asks for the **particular** solution – solve exactly as in Types 19-20 to reach the general solution with $+C$, then substitute the given point to find C .

★ **Board Favorite** – Linear DE with an initial condition is one of the most frequently repeated long-answer formats across ISC papers (2020, 2022, 2023 all carry a version of this exact combination).

Source: Exercise 9.7, Q20

Given: $(1 + x^2)\frac{dy}{dx} + 2xy - 4x^2 = 0, \quad y(0) = 0$

Solution:

Rearranging into standard linear form by moving $4x^2$ to the right and dividing by $(1 + x^2)$:

$$\frac{dy}{dx} + \frac{2x}{1 + x^2} y = \frac{4x^2}{1 + x^2}$$

This is the same standard form as Type 19’s worked example, and the same I.F. applies:

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1 + x^2$$

Multiplying through by the I.F. and integrating (left side collapses as usual):

$$y(1 + x^2) = \int 4x^2 \, dx = \frac{4x^3}{3} + C$$

Applying the initial condition $x = 0, y = 0$:

$$0 \cdot (1 + 0) = 0 + C \Rightarrow C = 0$$

Final Answer

$$y(1 + x^2) = \frac{4x^3}{3}$$

Alternate Board-Style Example

Source: Exercise 9.7, Q21, NCERT

Given: $\frac{dy}{dx} + y \cot x = 2x + x^2 \cot x, \quad y = 0 \text{ when } x = \frac{\pi}{2}$

Solution:

The equation is already in standard linear form with $P = \cot x$, so the Integrating Factor is:

$$\text{I.F.} = e^{\int \cot x \, dx} = e^{\log |\sin x|} = \sin x$$

Multiplying through by $\sin x$, the left side collapses to the derivative of $y \sin x$:

$$\frac{d}{dx} [y \sin x] = (2x + x^2 \cot x) \sin x = 2x \sin x + x^2 \cos x$$

Integrating the right side – notice $2x \sin x + x^2 \cos x$ is exactly the derivative (by the product rule) of $x^2 \sin x$, so this integrates directly:

$$y \sin x = x^2 \sin x + C$$

Applying the initial condition $x = \pi/2, y = 0$: $\sin(\pi/2) = 1$, so

$$0(1) = \left(\frac{\pi}{2}\right)^2 (1) + C \Rightarrow C = -\frac{\pi^2}{4}$$

$$y \sin x = x^2 \sin x - \frac{\pi^2}{4}$$

Dividing through by $\sin x$:

$$y = x^2 - \frac{\pi^2}{4} \csc x$$

Final Answer

$$y = x^2 - \frac{\pi^2}{4} \csc x$$

Quick Recall: Linear + initial condition: standard form, find I.F., collapse the left side, integrate the right side, then plug in the given point to solve for C – exactly the same steps as the general solution, one line longer.